National University of Computer and Emerging Sciences 0013
Islamabad Campus

Data Science Tools and Techniques (DS5002)

Course Instructor(s):
Dr. Muhammad Mateen Yaqoob
Section(s): MS DS

Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 50
Total Questions: 3

Date: Sep 23, 2025

251-8003 DS5002 DSTT
Roll No **Course Section** **Student Signature**

Do not write below this line.

Attempt all the questions.

DO NOT OPEN THE QUESTION PAPER OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt all the questions. Read the questions carefully, understand them, and then attempt it.
2. After being asked to commence the exam, please verify that you have **nine (09)** different printed pages and a total of **three (03)** questions.
3. No additional sheet will be provided for rough work. Use the provided answer-script for rough work.
4. Use permanent ink pens only. Any part done using a soft pencil will not be marked and cannot be claimed for rechecking.

The paper is a Closed-Book. You cannot use any electronic device.

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Q1: Consider the dataset below as given in table-1. **[15 marks]**

a. Apply the one-hot encoding and label-encoding techniques to the relevant column(s) of the above dataset and provide the results of both. Code is not required. (5 marks)

b. According to the results in (a), which encoding technique is better and why? (2 marks)

c. Identify the issues with the dataset. (3 marks)

d. Handle the issues identified in part (c) using relevant technique. Code is not required. (5 marks)

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Table-1: Financial dataset.

Age	Workclass	Education	Marital_Status	Occupation	Relationship	Race	Sex	Capital_Gain	Capital_Loss	Hours_per_Week	Native_Country	Income
39	State-gov	Bachelors	Never-married	?	Not-in-family	White	Male	2174	0	40	US	<=50K
50	Self-emp-not-inc	Bachelors	Married-civ-spouse	Exec-managerial	Husband	White	M	0	0	67	United-states	<=50K
38	Private	HS-grad	Divorced	Handlers-cleaners	Not-in-family	White	Male	0	0	40	United-State	<=50K
53	Private	11th	Married-civ-spouse	Handlers-cleaners	Husband	Black	Male	0	0	40	USA	<=50K
28	Private	Bachelors	Married-civ-spouse	Prof-specialty	Wife	Black	Female	0	0	40	Cuba	<=50K
37	?	Masters	Married-civ-spouse	?	Wife	White	Female	0	0	40	United-States	<=50K
49	Private	9th	Married-spouse-absent	Other-service	Not-in-family	Black	F	0	0	16	Jamaica	<=50K
Nan	Self-emp-inc	HS-grad	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0	45	America	>50K
31	Private	Masters	Never-married	Prof-specialty	Not-in-family	White	F	14084	0	50	United-States	>50K
42	Private	Bachelors	Married-civ-spouse	Exec-managerial	Husband	White	Male	5178	0	40	United-States	>50K
37	Private	Some-college	Married-civ-spouse	Exec-managerial	Husband	Black	Male	0	0	80	United-States	>50K
30	State-gov	Bachelors	Never-married	Protective-serv	?	White	Male	0	0	40	United-States of America	>50K
23	Private	HS-grad	Never-married	Adm-clerical	Own-child	White	Female	0	0	30	United-States	<=50K
32	Private	Assoc-acdm	Never-married	Sales	Not-in-family	Black	Nan	0	0	50	?	<=50K
40	Private	Assoc-voc	Married-civ-spouse	Craft-repair	Husband	Asian-Pac-Islander	Male	0	0	40	Philippines	>50K

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Q2: Consider the dataset given in table-2 below. **[15 marks]**

Table-2: Dataset from an online retail company.

Customer_ID	Name	Age	Gender	Income	Product_Category	Purchase_Amount	Country	Sales_Date
1001	john doe	29	M	45000	Electronics	500	USA	3/10/2022
1002	JANE SMITH	45	F	80000	electronics	1200	USA	5/11/2022
1003	Robert Brown	NaN	M	60000	clothing	850	usa	6/14/2022
1004	john doe	29	M	NaN	Furniture	300	Canada	7/1/2022
1005	LUCY LEWIS	35	Female	30000	furniture	1000	CANADA	7/15/2022
1006	Ann O'Brien	30	F	65000	Electronics	1200	UK	7/20/2022
1007	NaN	27	M	50000	Clothing	-50	United_Kingdom	7/25/2022
1008	Mike Davis	50	Male	40000	Elec	150	UK	15-08-2022
1009	Robert Brown	NaN	M	60000	Clothing	850	USA	9/10/2022
1010	lucy lewis	35	Female	30000	furniture	1000	USA	10/1/2022
1011	John Smith	42	M	55000	electronics	450	Usa	10/15/2022
1012	mary ann	38	female	70000	Furniture	900	usa	10/20/2022
1013	Bob Johnson	32	M	NaN	clothing	750	USA	10/25/2022
1014	ROBERT BROWN	NaN	Male	60000	Electronics	950	NaN	10/30/2022
1015	LUCY LEWIS	35	Female	30000	furniture	1000	CANADA	11/1/2022
1016	Ann O'Brien	30	F	65000	Electronics	1100	UK	11/5/2022
1017	JANE SMITH	45	F	NaN	electronics	1200	USA	11/10/2022
1018	Mike Davis	50	Male	40000	Elec	200	uk	11/11/2022
1019	Bob Johnson	32	M	80000	clothing	750	United States	11/15/2022
1020	john doe	29	M	45000	Electronics	500	USA	12/1/2022

a. Identify the data related issues in the above dataset. (2 marks).

b. Write Python code to overcome the issues identified in part (a). (5 marks)

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c. Write a code to compute the total purchase amount per country and sort the results in descending order. (3 marks)

d. For each customer, compute the total purchase amount. Write a code to identify the top 3 customers with the highest purchase amounts. (3 marks)

e. Find the number of purchases made per month across all years. Write a code to determine which month had the highest sales. (2 marks)

Q3: For the research paper below, answer the following. **[20 marks]**

a. How does their proposed CFCL differ from the existing frameworks? Why is a hierarchical interval-based approach particularly suitable for regression tasks such as blood pressure estimation?

b. The authors designed a Selection Layer (SL) with LASSO regularization. How does this differ from traditional feature selection methods?

c. What are the strengths and weaknesses of using LASSO for feature selection in physiological signals?

d. The authors evaluated performance on public, private, and synthetic datasets. Why is testing under domain shift important?

e. Do you think metrics like MAE and SD alone are sufficient to assess clinical reliability?

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expertise and extensive experimentation. Moreover, these features often exhibit interdependencies, further complicating selection. In DL methods, the heterogeneity of physiological vibration signals hinders generalization [13]. Contrastive learning can enhance representation distinctiveness and generalizability [6], but most existing approaches are designed for classification, emphasizing global features while overlooking crucial local semantics for regression. Addressing these limitations while integrating the strengths of both DL and DSP-ML remains a significant challenge.

In this paper, we propose a novel and effective framework for blood pressure estimation based on physiological vibration signals, denoted as CLFSS, which utilizes the coarse-to-fine Contrastive Learning, Feature Selection and Synthesis. Within the framework, we first design a seismometer-based vibration detector, BedDot, to collect vibration signals associated with cardiac activity for further processing. The BedDot system is illustrated in Fig 1. To learn robust representations with multiscale information, we innovate a Coarse-to-Fine Contrastive Learning (CFCL) module. It hierarchically classifies representations into progressively narrower intervals in a binary fashion and leverages the principles of contrastive learning at the same time. To leverage potentially relevant physiological features and overcome the tediousness of feature selection, an innovative Selection Layer (SL) is devised for feature selection, where we apply the Lasso regularization to a one-to-one connected layer and select features through the magnitude of the corresponding weights in this layer automatically. Following SL, a Multilayer Perceptron (MLP) is utilized to synthesize high-level features, thereby capturing complex interrelationships between selected features. Ultimately, to integrate the rich semantic information of learned representations and the stability inherent in DSP-derived features, we fuse the synthesized features with contrastive learning representations, based on which precise blood pressure estimation can be achieved.

2 Related Works

Physiological vibration signals are engendered by cardiac activities, which are intimately related to the state of the heart and thus can facilitate the estimation of blood pressure. With advantages like the contactless nature and no privacy implications, Seismocardiogram (SCG) and Ballistocardiogram (BCG) have gained increasing attention in recent years [18]. Das et al. observed that the AO and AC peaks of SCG can be used to estimate the approximate LVET that can facilitate blood pressure estimation [7]. Carlson et al. observed that the vertical and horizontal distance of IJ and JK of BCG can be used for blood pressure estimation [5].

In addition to the manual identification of features, deep learning is widely used to estimate blood pressure. However, the inherent heterogeneity of physiological vibration signals can result in a significant domain shift, posing challenges for the learned representations to generalize [2]. Contrastive learning is one of the domain generalization techniques to improve the discriminability and generalizability of representations by pulling representations of semantically similar samples together and pushing representations of samples with different semantic information apart [6]. In general, contrastive learning algorithms are tailored for classification tasks, which tend to capture global semantic information and overlook detailed local information to distinguish between closely related labels [24].

Feature selection improves model performance by selecting a subset of features and is categorized into Filter, Wrapper, and Embedded methods. Filters are efficient but ignore feature interactions [1], while Wrappers account for interactions but are computationally costly. Embedded methods like LASSO [21] and Ridge Regression [10] may oversimplify feature-label relationships or fail to explicitly select features. Feature synthesis combines base features to enhance model performance but often relies on predefined operations [12], limiting its ability to capture complex patterns.

3 Methodology

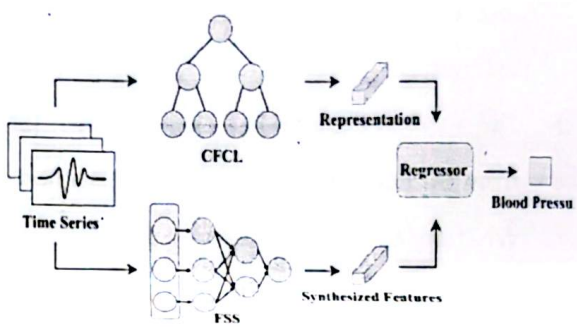


Figure 2: Overall Architecture of CLFSS

The proposed framework, referred to as CLFSS, consists of two parallel modules: the Coarse-to-Fine Contrastive Learning (CFCL) module and the Feature Selection and Synthesis (FSS) module. CFCL encodes input time series into latent representations, refining them through contrastive learning for improved generalizability. Meanwhile, FSS extracts DSP features, selects the optimal subset via the Selection Layer (SL), and synthesizes them using an MLP. Finally, CFCL representations and synthesized features are fused to enhance blood pressure estimation accuracy and robustness. The overall architecture of the CLFSS is depicted in Fig 2.

3.1 Coarse-to-Fine Contrastive Learning

To capture the generalizable and multiscale representations for regression tasks, we employ a coarse-to-fine contrastive learning (CFCL) module. It hierarchically classifies each representation into increasingly finer intervals in a binary manner. Through this process, contrastive learning is applied at each hierarchical level, treating representations within the same interval as positive pairs and others as negative pairs. The final representation is fused with synthesized features from the FSS module to estimate the blood pressure. The illustration of CFCL is shown in Fig 3.

3.1.1 Hierarchical Classification Given a time series dataset, $D = \{(x_1, y_1), (x_2, y_2), \cdots, (x_N, y_N)\}$, we employ an encoder $G(\cdot)$ to transform the time series $X = \{x_1, x_2, \cdots, x_N\}$ into latent representations $Z = \{z_1, z_2, \cdots, z_N\}$. Subsequently, the hierarchical and binary classification is performed to classify each z_i into corresponding intervals level by level. At the i-th level, an interval

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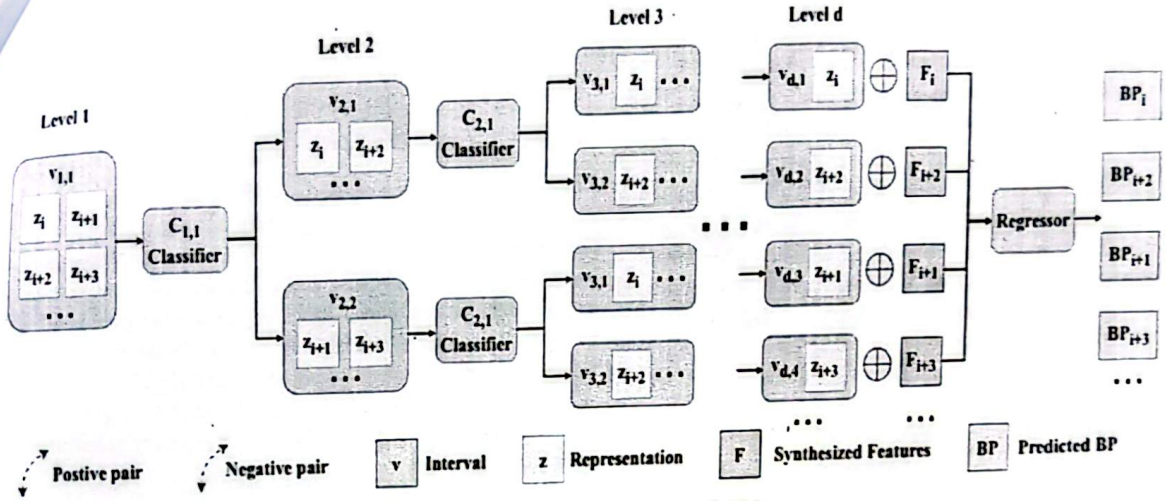


Figure 3: Overall architecture of CFCL

$v_{i,j} = [\zeta^l_{i,j}, \zeta^r_{i,j})$ represents a blood pressure range with the minimum value as $\zeta^l_{i,j}$ and maximum value as $\zeta^r_{i,j}$. It has two son intervals at the next level, which are $v_{i+1,2j-1} = [\zeta^l_{i,j}, (\zeta^l_{i,j} + \zeta^r_{i,j})/2)$ and $v_{i+1,2j} = [(\zeta^l_{i,j} + \zeta^r_{i,j})/2, \zeta^r_{i,j})$. The size of son intervals is half of that of their father interval and the size of final level's interval won't exceed 10 mmHg. The representations in $v_{i,j}$ will be classified into one of $v_{i+1,2j}$ and $v_{i+1,2j-1}$ by a trainable classifier $C_{i,j}(\cdot)$. If a representation is classified into $v_{i+1,2j}$, it indicates that the corresponding blood pressure range of the representation is $[(\zeta^l_{i,j} + \zeta^r_{i,j})/2, \zeta^r_{i,j})$.

Through this hierarchical classification process, the representation z is ultimately categorized into a certain interval at the last level, where the representation z is then integrated with the synthesized features produced by the FSS module to form the fused features $h = concat(z, F_s)$. Then a regressor $R(\cdot)$ performs the final blood pressure estimation based on h.

3.1.2 Contrastive Learning During the hierarchical classification process, we borrow the merits of contrastive learning to enhance the generalizability of representations and encourage them to capture the semantic information of different granularities.

For a given interval $v_{i,j}$, the representations within the interval will be classified into two son intervals $v_{i+1,2j-1}$ and $v_{i+1,2j}$ at the next level. Then a contrastive loss is computed by treating representations within the same interval as positive pairs and those in different intervals as negative pairs. Following [6], the contrastive loss for the n-th representation z_n within the interval $v_{i,j}$ can be derived as:

$$\mathcal{L}^{i,j,n}_{ctr} = -log \frac{\sum_p \mathbb{1}_{p \neq n} exp(sim(z_p, z_n))/\tau}{\sum_q exp(sim(z_q, z_n))/\tau} \quad (1)$$

where z_p denotes the representations in the same interval as z_n, and z_q in the denominator denotes the representations in different intervals from z_n.

In summary, the total loss for the CFCL module is the weighted sum of classification, regression, and contrastive losses. By optimizing the total loss, we aim to promote the ability of the representations to capture multiscale information, thus improving their generalizability and distinctiveness for blood pressure regression tasks.

3.2 Feature Selection and Synthesis

To extract, select, and synthesize the related features, the FSS module is introduced. It consists of three components: extractor, selector, and synthesizer, which correspond to TSFEL [3], Selection Layer (SL), and Multilayer Perceptron (MLP), respectively. Through training this module, taking the extracted feature set as input and the blood pressure as the target label, we can identify the optimal feature subset and synthesize them simultaneously. The synthesized features are subsequently integrated with the representations in the CFCL module to facilitate blood pressure estimation. The architecture of FSS is illustrated in Fig 4.

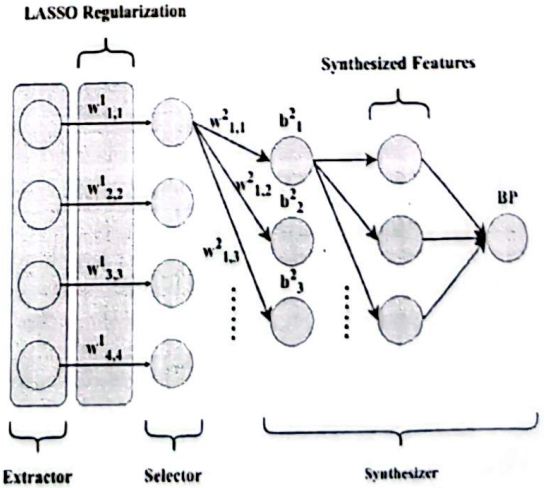


Figure 4: Overall architecture of FSS

First, we employ TSFEL [1] as the feature extractor $E(\cdot)$, an open source library for automatic time series feature extraction. It extracts common features from time, frequency, and statistic domains while allowing customization for specific physiological signals.

[1]Details about all the features extracted by TSFEL can be found in TSFEL feature_list

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Subsequently, we propose an innovative Selection Layer (SL) to perform feature selection by receiving the extracted feature vector as input. Unlike a traditional fully connected layer, the SL is a one-to-one connected layer without bias, where each neuron is exclusively linked to a single neuron within the input layer. To perform feature selection, LASSO regularization, also known as L1 regularization, is applied to the SL. Due to the gradient of the L1 norm remaining constant at either 1 or -1, the effect of regularization maintains consistent intensity until the weight is reduced to zero. As a result, through the training process, certain weights in the SL will converge to zero, effectively excluding the corresponding features from the subsequent computations and thus achieving the effect of feature selection.

Finally, a Multilayer Perceptron (MLP) will receive the output of SL and regress blood pressure. The output of the second-to-last layer of MLP is designated as the synthesized features. Since the SL filters out irrelevant features by assigning them negligible weights, it ensures that only highly related features are synthesized by the MLP.

By optimizing the parameters of SL and MLP during gradient descent, we can concurrently select the optimal feature subset and synthesize them, which will subsequently be integrated with the representations in the CFCL module to provide additional information for blood pressure estimation.

4 Experiments

4.1 Selection Layer

Scenario	Metric	MI	RFE	RF	SL
Linear	Recall	41.80	91.73	68.40	**96.13**
	Precision	9.85	7.29	51.17	**100.00**
Non-linear	Recall	41.40	**98.87**	68.67	89.47
	Precision	9.61	7.59	49.33	**97.25**
Interaction	Recall	39.13	96.53	73.80	**97.13**
	Precision	9.14	18.57	41.69	**96.36**
Synthetic SCG (SBP)	Recall	66.67	**100.00**	66.67	**100.00**
	Precision	15.38	9.09	28.57	**42.86**

Table 1: Feature selection results of SL

4.1.1 Experimental Setup To validate the Selection Layer (SL)'s ability to select the right features, we generated diverse equations simulating real-world feature-target relationships, including linear, non-linear, and interaction-based dependencies. A subset of features was randomly selected from a set of size 1000 to construct the dataset according to the equation. Additionally, synthetic SCG signals were used to assess SL's effectiveness in physiological vibration signal analysis, where blood pressure was modeled as a function of peak distances and heart rate. 560 time series features were also extracted using TSFEL to form the initial feature set. The purpose of the Selection Layer is to identify the correct features that are highly correlated with the labels from the initial feature set.

In the experiments, our SL-based feature selection was compared with three typical methods: Mutual Information (MI), Recursive Feature Elimination (RFE), and Random Forest (RF). Since MI and RF only provide feature importance rankings, thresholds (0.02 for MI, 0.002 for RF) were set via cross-validation. RFE, using lasso regression, iterated until the cross-validation score stopped improving, selecting the remaining features.

4.1.2 Analysis of Experimental Result We evaluated feature selection using Recall, Precision, and F1 Score, labeling right features as 1 and others as 0. As shown in Table 1, SL outperforms other methods across various categories. While RFE achieves similar Recall, its high computational cost makes it impractical compared with efficient SL. Additionally, SL's superior Precision prevents redundant feature selection, ensuring a balanced Recall and F1 Score. Overall, SL's strong performance on both generated functions and synthetic SCG features confirms its effectiveness in selecting relevant features.

4.2 Blood Pressure Estimation

4.2.1 Datasets To evaluate CLFSS for blood pressure estimation, we used three datasets: the public Bed-Based Ballistocardiography Dataset [4], our self-collected CTRU dataset, and a synthetic dataset. The public dataset includes 4.6 hours of data from 40 participants at 1000 Hz and the CTRU dataset contains 550 hours of data at 100 Hz. The latter one was collected by our BedDot and has been approved by the IRB. Quality control and bandpass filtering (3-14 Hz) were applied to remove noise. To test robustness under domain shifts, we generated a synthetic SCG dataset with label shift, where SBP ranges differed between training set and testing set. Each data sample was a 10-second signal, with 80% for training/validation and 20% for testing, ensuring no data leakage.

4.2.2 Experiment Setup Our proposed framework was compared with seven leading methods for time series analysis, including DL models such as the Multilevel Wavelet Decomposition Network (mWDN) [23], MiniRocket [20], Explainable Convolutional Neural Network (XCM) [8], InceptionTime [11] and Transformer [22]. We also manually extracted the features proposed by Su et al. [19], which were used as inputs for the ML models to compare with our proposed method. Furthermore, we modified the classic contrastive learning framework SimCLR [6] by combining it with supervised learning to compare it with our framework based on supervised learning. All methods underwent identical training, validation, and testing protocols. The estimation results were evaluated using Mean Absolute Error (MAE) and Standard Deviation (SD).

4.2.3 Analysis of Experimental Results Table 2 shows that our framework achieves the lowest MAE and SD for both SBP and DBP on all three datasets. In particular, on the synthetic SCG dataset with label shift for SBP, all baseline models suffered a significant decline in performance, which is attributable to their reliance on domain-specific features and poor generalization under domain shifts. In contrast, our framework enhances the generalizability of the representations through CFCL, and by concatenating the synthesized features with the representations, more stable and domain-invariant information is provided, thereby enhancing its ability to handle domain shifts. The FDA's ISO 81060-2:2018(E) requires a mean error within ±5.0 mmHg and a standard deviation no greater than 8.0 mmHg [14]. Our framework achieves mean errors of -1.34, 0.21, -1.98, and -1.15 mmHg for SBP and DBP on

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Model	SBP of BCG		DBP of BCG		SBP of CTRU		DBP of CTRU		SBP of Synthetic SCG	
	MAE	SD	MAE	SD	MAE	SD	MAE	SD	MAE	SD
mWDN	5.55	4.98	3.97	4.24	14.94	11.01	8.06	5.51	27.55	15.73
MiniRocket	5.47	4.70	4.42	3.75	12.97	10.71	6.82	5.46	10.35	7.92
XCM	5.51	4.79	4.26	4.12	15.87	12.37	8.15	6.33	42.12	16.19
InceptionTime	5.51	4.93	3.81	3.66	11.43	9.72	6.38	4.87	9.42	10.95
Transformer	7.92	6.13	6.68	5.23	15.27	13.13	8.11	7.04	16.27	12.70
SimCLR	5.31	4.78	3.92	3.68	11.14	10.42	6.22	5.86	17.99	14.58
Waveform Features	8.36	7.43	6.65	5.72	14.36	12.08	8.77	7.02	None	None
CLFSS (proposed)	**4.77**	**4.38**	**3.51**	**3.28**	**9.93**	**7.81**	**6.27**	**5.06**	**5.41**	**5.22**

Table 2: Blood pressure estimation results

public BCG and CTRU datasets, with standard deviations shown in Table 2. These results surpass FDA benchmarks, demonstrating the framework's feasibility for real-world application. To further validate the effectiveness of each module within the framework, we conducted ablation studies on the SBP of CTRU dataset. As shown in Table 3, it can be concluded that both the SL and CFCL modules improve the model performance, and when combined, they result in an obvious increase of 15.70% and 34.43% in MAE and ME relative to the baseline, which demonstrates the effectiveness of the two modules of our framework.

CFCL	FSS	MAE	SD	ME	MAPE (%)
✗	✗	11.78	10.39	-3.02	7.44
✓	✗	10.31	9.35	-2.57	7.12
✗	✓	11.15	9.89	-2.41	7.17
✓	✓	9.93	7.81	-1.98	6.72

Table 3: Ablation study on proposed modules

5 Conclusion

In this study, we propose an innovative blood pressure estimation framework via coarse-to-fine contrastive learning, feature selection and synthesis. The coarse-to-fine contrastive learning encourage the representations to capture the semantic information of different granularities. The novel SL and MLP are utilized to select and synthesize features, forming high-level features with stronger representational power. The experiments demonstrate the superiority of our algorithm in blood pressure estimation compared with other leading methods. In the future, we plan to conduct experiments in diverse environments and across different demographic groups to further validate and enhance its generalization performance.

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